

Pandas Library Methods - Complete Notes

1. Importing Data Methods

- `read_csv()`: Read CSV files
- `read_excel()`: Read Excel files
- `read_json()`: Read JSON files
- `read_sql()`: Read SQL databases
- `read_html()`: Read HTML tables
- `read_pickle()`: Read pickle files
- `read_parquet()`: Read parquet files

2. Creating DataFrame

- `DataFrame()`: Create DataFrame
- `Series()`: Create Series
- `from_dict()`: Create from dictionary
- `from_records()`: Create from records

3. Viewing Data

- `head()`: First rows
- `tail()`: Last rows
- `sample()`: Random rows
- `info()`: Data information
- `describe()`: Statistical summary
- `shape`: Rows and columns
- `size`: Total elements
- `ndim`: Dimensions
- `columns`: Column names
- `index`: Index values
- `dtypes`: Data types

4. Selecting Data

- loc[]: Select by label
- iloc[]: Select by position
- at[]: Single value by label
- iat[]: Single value by position
- filter(): Filter data
- query(): Query rows

5. Adding and Removing Data

- insert(): Add column
- assign(): Create columns
- drop(): Delete rows/columns
- pop(): Remove column

6. Data Cleaning

- isnull(), isna(): Find missing values
- notnull(): Check non-null values
- dropna(): Remove missing values
- fillna(): Fill missing values
- replace(): Replace values
- duplicated(): Find duplicates
- drop_duplicates(): Remove duplicates

7. Sorting

- sort_values(): Sort values
- sort_index(): Sort index
- nlargest(): Largest values
- nsmallest(): Smallest values

8. Mathematical Methods

- sum(), mean(), median(), mode()

- min(), max(), std(), var()
- count(), abs(), round()

9. Grouping

- groupby()
- agg()
- aggregate()
- transform()
- filter()

10. Combining Data

- concat()
- merge()
- join()

11. String Methods

- lower()
- upper()
- title()
- contains()
- replace()
- split()
- strip()
- startswith()
- endswith()

12. Date and Time

- to_datetime()
- date_range()
- dt.year

- dt.month
- dt.day
- dt.hour

13. Data Conversion

- astype()
- convert_dtypes()
- to_numpy()
- to_dict()
- to_string()

14. Reshaping Data

- pivot()
- pivot_table()
- melt()
- stack()
- unstack()
- transpose()

15. Export Methods

- to_csv()
- to_excel()
- to_json()
- to_sql()
- to_html()
- to_pickle()

16. Statistical Methods

- corr()
- cov()
- rank()

- quantile()

- skew()

- kurt()

17. Advanced DataFrame Methods

- apply()

- applymap()

- map()

- copy()

- rename()

- reset_index()

- set_index()

- reindex()

- memory_usage()

18. Visualization

- plot()

- hist()

- boxplot()

Most Important Pandas Methods:

read_csv(), head(), tail(), info(), describe(), loc[], iloc[],

drop(), fillna(), dropna(), groupby(), merge(), concat(),

sort_values(), value_counts(), unique(), apply(), map(),

astype(), to_datetime(), pivot_table(), corr(), plot(), to_csv()